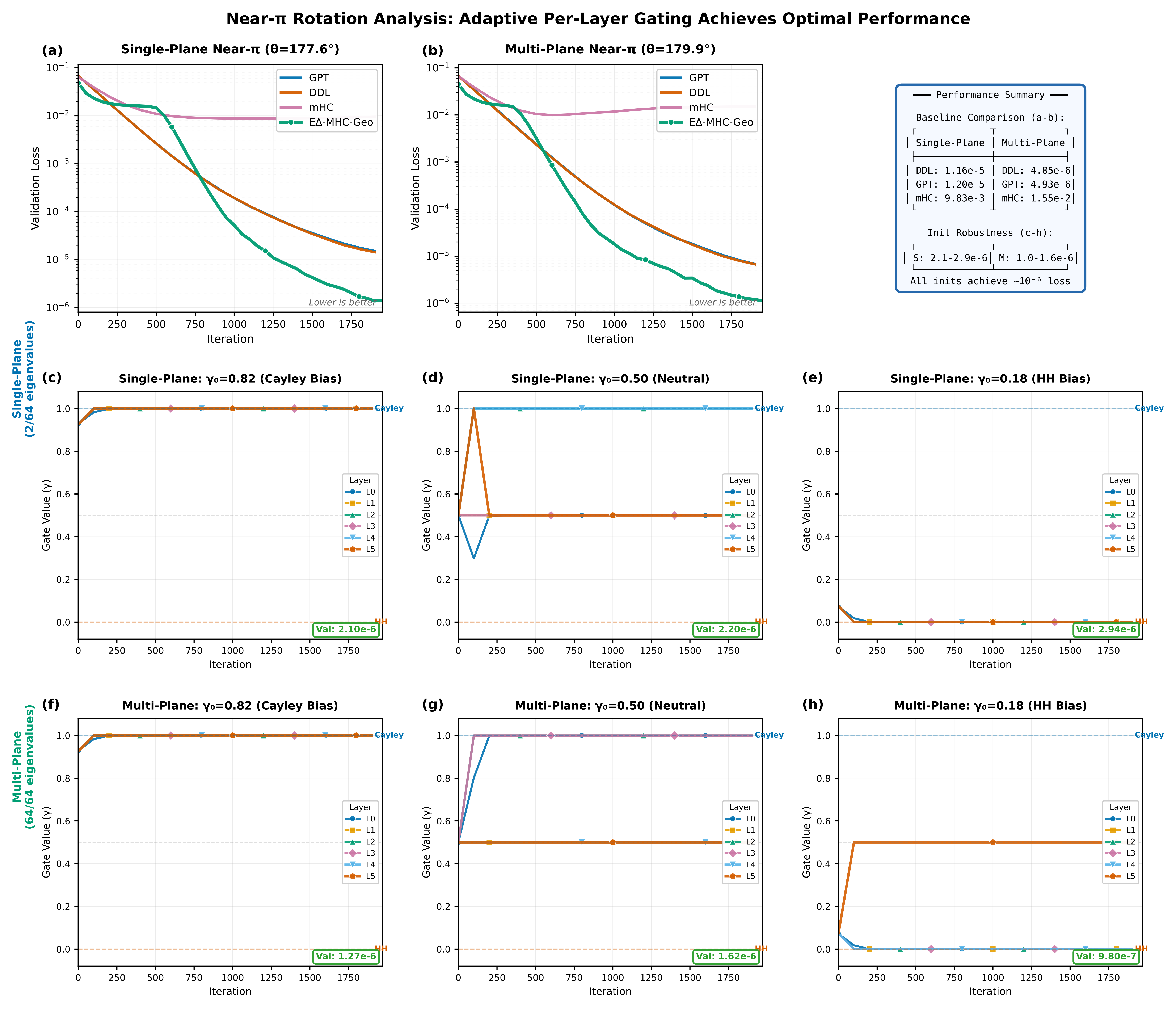
Near-π Rotation Analysis: Adaptive Per-Layer Gating Achieves Optimal Performance
(a) Single-Plane Near-π (θ=177.6°)
(b) Multi-Plane Near-π (θ=179.9°)
GPT
DDL
mHC
EΔ-MHC-Geo
Validation Loss
Iteration
Lower is better
Performance Summary
Baseline Comparison (a-b):
Single-Plane | Multi-Plane
DDL: 1.16e-5 | DDL: 4.85e-6
GPT: 1.20e-5 | GPT: 4.93e-6
mHC: 9.83e-3 | mHC: 1.55e-2
Init Robustness (c-h):
S: 2.1-2.9e-6 | M: 1.0-1.6e-6
All inits achieve ~10⁻⁶ loss
Single-Plane (2/64 eigenvalues)
Multi-Plane (64/64 eigenvalues)
(c) Single-Plane: γ₀=0.82 (Cayley Bias)
(d) Single-Plane: γ₀=0.50 (Neutral)
(e) Single-Plane: γ₀=0.18 (HH Bias)
(f) Multi-Plane: γ₀=0.82 (Cayley Bias)
(g) Multi-Plane: γ₀=0.50 (Neutral)
(h) Multi-Plane: γ₀=0.18 (HH Bias)
Gate Value (γ)
Layer
L0
L1
L2
L3
L4
L5
Cayley
HH
Val: 2.10e-6
Val: 2.20e-6
Val: 2.94e-6
Val: 1.27e-6
Val: 1.62e-6
Val: 9.80e-7